

Design Thinking (AECC27)

UNIT-V: Case Studies

1. What is data-driven design? How to use data in your design process? Explain the above with the help of a relevant case study example.

Data-driven design is an approach to designing products, services, or experiences that relies on data analysis and insights to inform decision-making throughout the design process. It involves collecting, analyzing, and interpreting data about user behavior, preferences, and needs to create more effective and user-centric designs.

Here's how you can incorporate data into your design process:

- a) **Define Goals and Metrics:** Clearly define your design goals and the key metrics you'll use to measure success. These metrics could include user engagement, conversion rates, task completion times, etc.
- b) **Collect Data:** Gather relevant data from various sources such as user surveys, interviews, usability tests, A/B testing, and user feedback.
- c) **Analyze Data:** Use statistical methods and data analysis techniques to make sense of the collected data. Identify patterns, trends, and insights that can guide your design decisions.
- d) **Monitor and Measure:** Continuously monitor the performance of your design using the defined metrics. Track how users interact with the design and make adjustments as needed based on ongoing data analysis.
- e) **Iterative Improvement:** Data-driven design is an iterative process. Use the data collected over time to refine and improve your design continually. Regularly revisit your design goals and metrics to ensure alignment with user needs and business objectives.

Case study example:

Here are some key points about Amazon's data-driven design approach for its e-commerce website:

- i. **Personalized Recommendations:** Amazon uses data analysis to offer personalized product recommendations based on users' browsing and purchase history.
- ii. **Dynamic Pricing:** They employ algorithms to adjust prices dynamically based on factors like demand, competitor pricing, and user behavior.
- iii. **User Reviews and Ratings:** Amazon prominently displays user reviews and ratings for products, helping shoppers make informed purchasing decisions.
- iv. **One-Click Purchase:** They pioneered the one-click purchase feature, streamlining the checkout process based on user data to reduce friction.
- v. **Predictive Search:** Amazon's search functionality predicts and suggests products as users type, improving search accuracy and speed based on historical search data.
- vi. **Supply Chain Optimization:** Data analysis helps optimize inventory management, predicting demand and ensuring products are available when and where customers need them.

2. How do you create a good design thinking project? Explain.

Creating a good design thinking project involves several key steps to ensure that the project is well-defined, addresses real user needs, and fosters innovation. Here's a step-by-step guide to creating a good design thinking project:

1. Identify the Problem or Opportunity: Start by identifying a specific problem or opportunity that you want to address through design thinking. This could be a pain point experienced by users, a business challenge, or an area for innovation.

2. Define the Scope: Clearly define the scope of the project, including its objectives, target audience, and any constraints or limitations. Make sure the scope is realistic and achievable within the resources available.

- 3. Research and Empathize:** Conduct research to gain a deep understanding of the problem or opportunity from the perspective of users and stakeholders. Use methods such as user interviews, surveys, observations, and empathy mapping to gather insights and empathize with the people affected by the problem.
- 4. Generate Ideas:** Facilitate brainstorming sessions or ideation workshops to generate a wide range of ideas for addressing the problem or opportunity. Encourage creativity and collaboration among team members to come up with innovative solutions.
- 5. Prototype Solutions:** Select the most promising ideas and create prototypes to visualize and test potential solutions. Prototypes can range from low-fidelity sketches or wireframes to high-fidelity mockups or prototypes, depending on the complexity of the project.
- 6. Test and Iterate:** Test the prototypes with real users to gather feedback and insights. Use methods such as usability testing, user interviews, and prototype testing to identify strengths, weaknesses, and areas for improvement. Iterate on the prototypes based on user feedback, refining and optimizing the solutions.
- 7. Implement and Scale:** Once you have validated and refined the solutions through testing, move forward with implementing them. Develop an implementation plan and timeline, and work closely with stakeholders to ensure a smooth rollout. Monitor the implementation process and make adjustments as needed.
- 8. Evaluate and Learn:** After the solutions have been implemented, evaluate their effectiveness against the project objectives and metrics. Collect feedback from users and stakeholders to understand how well the solutions are meeting their needs. Use this feedback to learn from the project and identify opportunities for further improvement or iteration.

3. What are some of the tips for idea generation and synthesis with large groups?

Generating and synthesizing ideas with large groups can be challenging but also highly rewarding in terms of creativity and innovation. Here are some tips for idea generation and synthesis with large groups:

- i. **Create a Diverse Group:** Ensure that the group is diverse in terms of backgrounds, expertise, and perspectives. Diversity can lead to a broader range of ideas and more innovative solutions.
- ii. **Set Clear Objectives:** Clearly communicate the goals and objectives of the ideation session. Make sure everyone understands the problem or opportunity you're addressing and what you hope to achieve.
- iii. **Provide Inspiration:** Start the session by providing inspiration or examples related to the problem or opportunity. This could include case studies, success stories, or examples of innovative solutions from other industries.
- iv. **Use Divergent Thinking Techniques:** Encourage divergent thinking by using brainstorming techniques such as mind mapping, SCAMPER (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse), or the "Yes, and..." approach from improv theater.
- v. **Encourage Collaboration:** Encourage collaboration and build on each other's ideas. Use techniques such as idea clustering or building on previous ideas to generate new insights and concepts.
- vi. **Set Time Limits:** Set time limits for each ideation exercise to keep the energy level high and prevent discussions from dragging on. Timeboxing can also encourage participants to focus on generating ideas quickly.
- vii. **Synthesize Ideas:** After the ideation phase, facilitate a synthesis process to review and prioritize the ideas generated. Use techniques such as affinity mapping or voting to identify the most promising ideas and create a plan for further development.

Case Studies:

1. Amul Case Study:

Amul, the dairy company, strategically navigated through the COVID-19 pandemic by increasing production, procuring more milk, and utilizing digital technology. They capitalized on the change in consumer behavior and invested heavily in marketing to increase their revenue and support farmers.

Key Insights:

- i. **Strategic Preparation:** Amul took a bold decision to operate at full capacity during the pandemic, leveraging consumer behavior changes and market potential.
- ii. **Effective Supply Chain Management:** Amul's partnership with IBM and digital system allowed them to track and optimize their extensive supply chain network.
- iii. **Marketing Resilience:** Amul increased their advertising volume during the pandemic, creating brand visibility and capitalizing on nostalgic marketing campaigns.
- iv. **Incentivizing Workers:** Amul provided incentives and support to their frontline workers, building strong relationships and ensuring operational efficiency.
- v. **Lessons for Business Leaders:** Great leaders embrace risk, prioritize resourcefulness, and see crises as opportunities to get ahead of competitors.
- vi. **Capitalizing on Market Potential:** Amul tapped into the changing consumer behavior and increased demand for milk products, gaining market share and supporting farmers.

Importance of Marketing: Despite the pandemic, Amul's aggressive marketing efforts helped them maintain brand visibility and increase revenue.

2. Asian Paints Case Study:

Asian Paints is a leading company in the Indian paint industry, growing at a remarkable rate. It has consistently outperformed competitors in terms of revenue and profit generation. The company's success can be attributed to its market research, innovative products, efficient supply chain, and strong relationships with dealers.

Key Insights

- i. Asian Paints' success lies in its ability to identify and cater to different market segments. By understanding the industrial and decorative paint segments, the company was able to capture both B2B and B2C markets.
- ii. The company's efficient supply chain and superior technology have played a significant role in its success. By leveraging technology, Asian Paints has improved efficiency, reduced working capital requirements, and achieved remarkable growth.
- iii. Asian Paints' strong relationships with dealers have been crucial to its market dominance. By offering incentives for regular payments and providing support and financing options, the company has built trust and loyalty among its dealers.
- iv. Asian Paints' focus on continuous improvement and adoption of new technologies, such as mainframe computers and GPS, has allowed the company to stay ahead of the competition and operate at unprecedented levels of efficiency.
- v. The company's consistent market leadership for over five decades is a result of its ability to adapt to changing market dynamics, invest in research and development, and deliver superior products and services to its customers.
- vi. Asian Paints' success story serves as a valuable lesson for businesses, highlighting the importance of market research, efficient supply chain management, strong dealer relationships, and innovation in product development and marketing strategies.

3. Futuristic Farms Case Study:

This case study explores the Netherlands' approach to sustainable agriculture, which has made them the world's second-largest food exporter despite having limited land.

Key points:

The challenge: The world's food system faces immense challenges, including a growing population, changing consumption patterns, and climate change. To meet the needs of the future, we need to produce more food with fewer resources.

The Dutch solution: The Netherlands has embraced sustainable agriculture practices that allow them to produce twice as much food with half the resources. This is achieved through a combination of factors, including:

- i. **Collaboration:** There is close collaboration between the government, science organizations, and industry, all working towards the shared goal of sustainable food production.
- ii. **Innovation:** The Dutch are constantly innovating new technologies and methods to improve efficiency, such as using high-tech greenhouses, LED lights, and AI-powered climate control systems.
- iii. **Knowledge sharing:** The Netherlands is committed to sharing its knowledge and expertise with other countries around the world.

Outcome:

- i. **Tomato production:** In the Netherlands, greenhouses allow farmers to produce 80 kilograms of tomatoes per square meter, compared to just 4 kilograms in open fields. This is achieved through a variety of techniques, including using LED lights and controlling the climate.
- ii. **Pest control:** The Dutch are using drones to kill moths, which are harmful pests for many crops. This is a more efficient and sustainable alternative to traditional pesticides.
- iii. **Climate control:** Some Dutch greenhouses are using AI to learn plant behavior and adjust conditions automatically, without any input from a farmer.

The case study concludes by suggesting that the Netherlands' approach to sustainable agriculture can serve as a model for other countries to follow in order to address the global food challenge.

4. Data Driven Design Case Study:

This case study is about how data will be one of the most important design trends of the 2020s and beyond. Companies like Microsoft, Nike, and Xaia have used data to inform product design decisions. However, there are ethical controversies around data collection and how some types of data are much more valuable than others.

One of the main points is that data can be used to create more comfortable and effective products. For example, Microsoft used data on thousands of people's heads to design the HoloLens 2, a headset that is more comfortable than previous models. Nike is using data to determine the outsole structure and pattern for some of their highest performing shoes.

Data can also be used to personalize products for individual users. For example, Nike made a customized outsole for each of their athletes' unique foot shapes and running event, and it helped their team win 45 medals during the 2016 Rio Olympics.

However, there are ethical concerns around data collection. Companies need to be careful about how they collect data and how they use it. They should only collect data that is necessary and they should be transparent with users about what data is being collected and how it is being used.

Overall, the data can be used to improve product design. However, it is important to be aware of the ethical concerns around data collection.

5. Redesigning the customer contact center at Toyota

Toyota's customer contact center faced challenges, including long wait times, repetitive inquiries, and frustrated customers. To address these issues, they embarked on a customer-centric redesign using design thinking:

- i. **Identifying the problem:** Long wait times, repetitive questions, and customer frustration were identified as key issues.
- ii. **Understanding the user:** They conducted research to understand both customer and call center representative perspectives.
- iii. **Developing solutions:** Based on their findings, they implemented several changes:
- iv. **Improved self-service options:** An online knowledge base and FAQs section were established to empower customers to find answers independently.
- v. **Streamlined call routing:** Incoming calls were directed to specialists equipped to handle specific inquiries, reducing wait times and improving resolution rates.
- vi. **Empowered representatives:** Representatives were given access to better resources and training, allowing them to address customer concerns more effectively and efficiently.
- vii. **Continuous improvement:** A feedback loop was established to gather customer and representative input, enabling ongoing refinement of the system.

Outcomes:

- a. **Reduced wait times:** Customers experienced shorter hold times, leading to a more positive experience.
- b. **Increased efficiency:** Representatives handled calls more efficiently, freeing up time to address complex customer needs.
- c. **Improved customer satisfaction:** Customer satisfaction with the contact center significantly increased.

This case study demonstrates the power of design thinking in customer service. By focusing on customer needs and experiences, Toyota successfully redesigned their contact center, improving both customer satisfaction and operational efficiency.

6. Rethinking Subsidized meals for the Elderly at The Good Kitchen

The Good Kitchen, a program delivering subsidized meals to the elderly in Denmark, faced challenges. Seniors felt a loss of control and social connection, while kitchen staff lacked motivation. To address this, designers used design thinking, a human-centered approach, to:

Gather insights: They held workshops with seniors, staff, and stakeholders to understand their needs and concerns.

Identify problems: They discovered seniors desired control, connection, and familiar flavors, while staff felt undervalued and uninvolved.

Develop solutions: Through collaborative efforts, they proposed changes like:

Choice and customization: Offering options for portion sizes and side dishes.

Improved communication: Sharing menus in advance and allowing feedback.

Meaningful work for staff: Involving them in menu development and interacting with seniors.

These changes aimed to improve the dining experience and well-being of both seniors and staff, creating a more dignified and engaging program.

This potentially addressed malnutrition by increasing food intake and improving dietary quality. Seniors were given more control over their food choices, which could lead them to select more nutritious options that they are more likely to enjoy and eat. Additionally, improved communication with staff could ensure that the meals provided are culturally appropriate and meet any specific dietary needs. Overall, the changes implemented by The Good Kitchen have the potential to improve the nutritional status of the elderly population they serve.